

An Efficient Approach for Encrypted Traffic Classification using CNN and Bidirectional GRU

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Abstract—With the rapid development of the Internet, the amount and types of network traffic have increased dramatically in tandem. Therefore, precise network traffic classification has become an essential aspect of network management. Furthermore, as user privacy and data encryption requirements have grown, more encrypted traffic has evolved. Because of the wide variety of encryption techniques and methodologies, network traffic categorization has become extremely challenging. Traditional traffic classification methods have been unable to meet the demand for classification accuracy. In this study, we propose a network structure that combines CNN and Bi-GRU to learn the temporal and spatial features of encrypted traffic data. We utilize the public dataset ISCX VPN-nonVPN to evaluate the effect of our model, and we use accuracy, recall, and F1 score as criteria. Finally, our model had a classification accuracy of 93.1%, with a recall rate of 93.7% and an F1 score of 93.6%. We also discussed about the differences between Bi-GRU and LSTM in terms of model parameter scale and time efficiency. Experiments show that Bi-GRU has the greatest classification effect and efficiency.

Keywords: Deep Learning; Network Traffic Classification; CNN; Bi-GRU

I. INTRODUCTION

With the rapid development of Internet communication technology in recent years, the network has become an integral element of people's work and lives. People's requirements have gotten more diverse and varied as the Internet has grown, resulting in additional application situations and developmental direction. The perception and monitoring of Internet traffic has become increasingly critical in such an environment. In today's network management system, network traffic categorization is a critical responsibility. The main goal is to predict the network data flow protocol and application type. Traffic categorization activities are crucial to network management and service quality with the fast development of high-throughput traffic demand. Hence, network traffic classification has aroused significant interest in academia and industry, and an increasing number of academics are paying attention to it.

In recent years, with the fast development in demand for protecting transmission data and user privacy, an increasing number of apps and protocols have begun to send data using encryption technologies, and the share of encrypted traffic in the network has grown dramatically. Various encryption mechanisms are now in use on the network. SSH, IPsec, TLS, and other encryption protocols were introduced in [1]. It is challenging to create a generic categorization system due to the variances in protocols and encryption techniques. The categorization of traffic poses significant difficulties. Traditional traffic categorization approaches such as port-based and deep packet inspection can't tackle this sort of problem well since encrypted communication is no longer in plaintext. At the moment, machine learning methods are primarily used to solve the problem of encrypted traffic classification, but this type of method requires the extraction of characteristics from network data streams, which means that developing an encrypted traffic classification system based on machine learning requires a significant amount of labor and domain knowledge. Migration and large-scale applications are tough to achieve. It is vital to re-extract features and train classifiers as new traffic kinds occur. Furthermore, owing to the limits of the machine learning model, the quality of the picked features typically impacts the classifier's ultimate effect, resulting in low classification accuracy. Deep learning, on the other hand, has advanced significantly in recent years, and an increasing number of academics have successfully implemented it in their specialties. This has also resulted in fresh concepts and solutions for the problem of classified encrypted traffic.

Therefore, this study provides a deep learning-based solution for encrypting traffic categorization. The network employs a hybrid paradigm that combines CNN and Bi-GRU. A bidirectional GRU network is known as Bi-GRU. The spatial features of data are learned using a one-dimensional CNN. GRU, being a kind of recurrent neural network, inherits the benefits of LSTM while also having a simpler structure, making it more effective in network training. The network can learn the forward and backward features, and our model can learn the

spatial and temporal properties of the traffic data, by using Bi-GRU to receive the output after convolution.

The remainder of this article is structured in the following manner. Section II discusses related work. Section III describes the design and implementation of the proposed method. Section IV mainly covers evaluation methods and experimental results. Section V presents conclusions and future work.

II. RELATED WORK

The existing approaches for encrypted traffic classification are classified from several viewpoints of the characteristics and models. Machine learning-based approaches and deep learning-based methods are the most often utilized methods nowadays.

A. Machine learning based methods

Velen et al. [1] studied papers on the classification of encrypted traffic between 2005 and 2015, and the research shows that before deep learning is widely used, payload-based and feature-based network classification methods are the main methods of encrypted traffic classification. Sun et al. [2] proposed an encrypted traffic classification method based on the naive bayes method. They used a hybrid method based on signature and statistical analysis and achieved a classification accuracy of more than 99%. Arndt [3] examined the impact of C4.5, k-means, and a multi-objective genetic algorithm on network traffic categorization. According to the study, C4.5 performs better in the categorization effect. In [4, 5], the support vector machine is used to classify applications. Among these methods, the feature selection method requires manual extraction of features for each category. This method necessitates extensive prior knowledge of the area, resulting in the model's weak generalization skills and inability to migrate to new settings.

B. Deep learning based methods

With the rapid development of deep learning, researchers have started to apply deep learning technology to implement encrypted traffic classification. Rezaei et al. [6] summarized different models for network traffic classification, of which convolutional neural networks are the most widely used. Wang et al. [7] was the first to apply CNN to network traffic classification. They proposed different traffic representation methods, among which session and all layers achieved the best results. Besides, they proposed using one-dimensional convolution neural networks to create an end-to-end encrypted traffic classification method [8]. They have an accuracy rate of 86.6% in the public dataset ISCX VPN-nonVPN [9] traffic classification. They also proposed in [10] that the HAST-IDS algorithm combined with CNN and LSTM be used to the DARPA1998 and ISCX2012 datasets, with 99 percent accuracy. Lotfollahi et al. [11] compares the classification results of one-dimensional CNN and stacked auto-encoders (SAEs), and experiments show that one-dimensional convolution is better. In [12, 13], they all advocated using an attention mechanism and a recurrent neural network to achieve traffic categorization. Although CNN is commonly used in these deep learning approaches, but CNN may not be able to learn the data's temporal properties. LSTM has succeeded with sequence data, however due to the complexity of the classification problem, it may be restricted by memory. Therefore, a lightweight and

efficient network is critical, this research offers a network model based on a CNN and Bi-GRU combination.

III. METHODOLOGY

A. Dataset

Owing to the majority of the dataset utilized in the articles are inconsistent, they often employ private dataset for network traffic categorization tasks. As a result, utilizing universal criteria to compare models and algorithms is problematic. Therefore, we used the publicly available ISCX VPN-nonVPN dataset, which was generated by the University of New Brunswick team. This dataset includes 26GB of encrypted traffic data in pcap and pcapng formats. The flow features of the dataset may be split into 14 categories, according to their paper's description, including Browse, Email, Chat and so on. However, due to the dataset's homogeneity, it's hard to tell the difference between the Browser and Streaming categories. Finally, we used [8]'s approach to divide the traffic data into 12 categories, six of which are VPN categories and others are not. Table I lists the specific category and size information.

TABLE I. ISCX VPN-NONVPN DATASET

Traffic Type	Content	Size
Chat	ICQ, AIM, Skype, Facebook and Hangouts	29.5M
VPN-Chat		27.6M
Email	Email, Gmail (SMTP, POP3, IMAP)	13M
VPN-Email		7.8M
File	Skype, FTPS and SFTP	17.3G
VPN-File		279M
Streaming	Vimeo, Youtube, Netflix, Spotify	1.53G
VPN-Streaming		1.37G
VoIP	Facebook, Skype and Hangouts voice calls	4.48G
VPN-VoIP		360M
Torrent	uTorrent, Bittorrent	96.8M
VPN-Torrent		358M

B. CNN Model

Convolutional neural networks have had a lot of success in a variety of domains, including speech recognition, picture recognition, natural language processing and others. CNN extracts feature by convolution and then lowers the order of magnitude of network parameters through convolutional weight sharing and pooling. CNN bypasses the difficult feature extraction method since it can successfully learn features from a large amount of data. Considering network traffic data is structured sequence data, we use CNN as part of our model to learn the spatial properties of the traffic data. The CNN hierarchy is as follows:

1) *Input*: Let x_i be the L-dimensional data of one of the session data including n packets, where $\oplus$ is the concatenation operator, and input data x may be written in series as $x_1, x_2, \dots, x_n$.

$$x_{1:n} = x_1 \oplus x_2 \oplus \dots \oplus x_n \quad (1)$$

2) *Convolution operation*: The convolution operation actually applies the convolution kernel w to the sequence data. Where w denotes the weight of the h-byte traffic data and b denotes the deviation term.

$$c_i = f(w \cdot x_{i:i+h-1} + b) \quad (2)$$

The final feature map formed after the convolution kernel passes through each flow data window is:

$$c = [c_1, c_2, \dots, c_{n-h+1}] \quad (3)$$

3) *Pooling operation*: The pooling operation can minimize the dimensionality of the feature map, reducing the amount of parameters ultimately transferred to the fully connected layer and increasing computation performance while avoiding overfitting. Finally, we use the maximum pooling approach.

$$\hat{c} = \max \{c\} \quad (4)$$

C. GRU Model

Recurrent neural networks are mostly used to learn the temporal properties of network traffic sequences. LSTM [14] and GRU [15] are two extensively utilized recurrent neural networks. To simulate long-term context and other interactions, LSTM employs a gating mechanism that allows the recurrent neural network to not only recall prior information, but also to selectively forget certain unimportant information. GRU may be thought of as an LSTM. It is a version that addresses the issue of disappearing gradients while preserving long-term sequence information. More than that, the parameters are less and training time of GRU during training are quicker than LSTM, which is why we use GRU as part of our network.

1) *Update gate*: The update gate assists the model in determining how much information from the past should be given to the future, or how much information from the previous and current time steps should be passed on. This is quite powerful since the model may choose to replicate all previous information to minimise the danger of disappearing gradients.

$$Z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1}) \quad (5)$$

Where x_t is the t-th time step's input vector, that is, the t-th component of the input sequence x that will be linearly transformed (multiply by the weight matrix $W^{(z)}$). h_{t-1} preserves information from the previous time step $t-1$ and also performs the linear transformation. The update gate combines these two pieces of information and feeds them into the sigmoid activation function, compressing the activation result to a value between 0 and 1.

2) *Reset gate*: The reset gate primarily decides how much past information must be erased.

$$R_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1}) \quad (6)$$

The output will employ the reset gate to save important information in the past, and its calculation expression is:

$$h'_t = \tanh(Wx_t + r_t \odot Uh_{t-1}) \quad (7)$$

where $\odot$ represents an element-wise multiplication symbol.

The outputs of the update gate and the reset gate must be mixed in the final phase.

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t \quad (8)$$

The state of a unidirectional neural network structure is always output from front to back. Bidirectional GRU is a neural network model made up of unidirectional and opposing GRUs, the output of which is dictated by the states of these two GRUs. A framework like this can collect more detailed and precise information. Therefore, our network employs a bidirectional GRU. Assuming $\vec{h}_t$ is the forward output of GRU and $\overleftarrow{h}_t$ is the backward output of GRU, then BiGRU's output may be represented as:

$$h_t = [\vec{h}_t, \overleftarrow{h}_t] \quad (9)$$

D. Network Structure

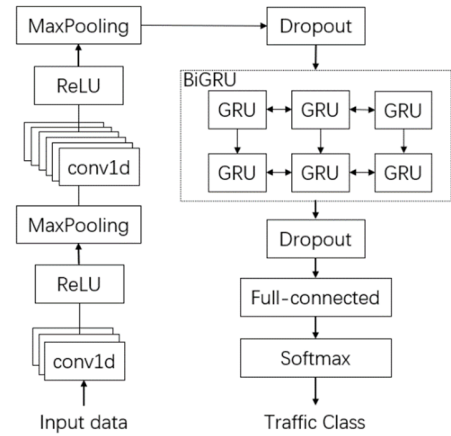


Figure 1. Network structure

Given that network traffic data is organized and has temporal characteristics, we perform network traffic classification tasks using a framework that combines convolutional neural networks and recurrent neural networks. Figure 1 depicts our neural network structure diagram.

As shown in Figure 1, the input data is initially sent through a one-dimensional convolutional neural network. The convolutional layer has 32 filters, a convolution kernel width of 3, a step size of 1, and padding of 1. The next convolution layer has 64 filters, a convolution kernel with a step size of 2, and the rest is the same as the first layer of convolution. The ReLU activation function is present in all convolutional layers.

Because each data packet's dimensionality is quite high, each convolution is followed by a maximum pooling operation with a width of 2 and a step size of 2 to minimize the data dimensionality. The spatial properties of the data are retrieved after convolution. The network data is context-sensitive since it has temporal properties. Bi-GRU is a network structure that learns the timing characteristics of data packets in order to determine the forward direction of network traffic as well as backward information. Bi-GRU learns temporal characteristics, transfers them to a fully connected network, and then outputs the final category through a Softmax layer. It's worth mentioning that after the convolution and recurrent neural networks, a dropout layer is linked, with a dropout rate of 0.2. This step prevents the network from being overfit.

IV. EXPERIMENT

A. Data Preprocessing

Data preprocessing is necessary for the ISCX VPN non-VPN dataset since it comprises the actual flow data, which is kept in packets. The following are the particular steps:

1) *Data segmentation*: We separate the original flow data and the huge data file into little files based on session in this stage. The flow data is segmented by source port, source IP, destination port, destination IP, and transport layer protocol. However, flow data is only in one direction, but flow data in both directions is included in the session. As a consequence, the session has additional details about both persons involved in the contact. The raw traffic data comprises 310457 session data once data segmentation is done.

2) *Remove useless information*: Each data packet is made up of TCP/IP protocol, and the data link layer information isn't particularly useful for later categorization, thus the data link layer data can be removed. HTTPS and QUIC are the most often used protocols for encrypted transmission. There are also other irrelevant data packets that should be eliminated, such as DNS protocol and others. Furthermore, because the dataset will employ numerous fixed IP addresses or ports to gather related category data during the collection process, it is extremely possible that one IP will collect a specific category of data, which will introduce additional information into the model that must be deleted. The number of session data is shown in Figure 2. After eliminating unnecessary data packets, there are 46,986 session data in total.

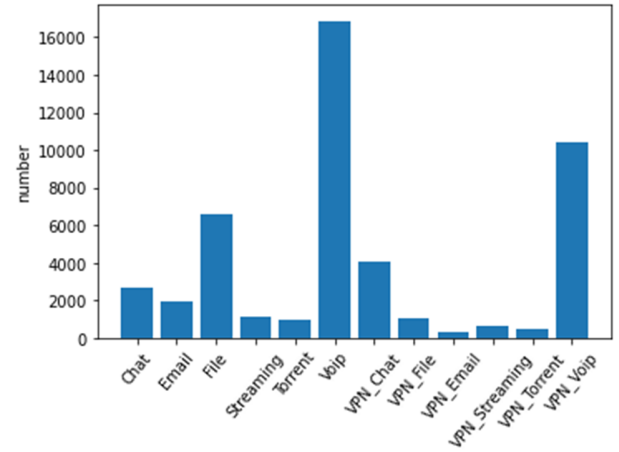


Figure 2. The number of different classes of data

3) *Data representation*: The data packet is made up of a byte array with a data range of $[0, 255]$ for each byte. As a result, we should first normalize these values and translate them to the range $[0, 1]$. After finishing data normalization, we transform the data into a $N * L$ matrix format, where N indicates the number of data packets contained in each session, and L represents each packet's first L bytes of data. If a data packet's length is less than L bytes, the data must be filled with 0 to L bytes. If the data packet's length exceeds L bytes, the surplus data must be trimmed.

B. Evaluation

1) *Experiment environment*: PyTorch was chosen as the deep learning framework for our experiment since it enables GPU execution and is easier to implement. As a result, PyTorch is used for all of the training and testing in this experiment. The experimental environment is Windows 10 version 20H2 with an AMD Ryzen 5 3600 3.60GHz CPU, 16GB of RAM, and an Nvidia GeForce GTX 1650 graphics card (4GB).

The ISCX VPN-nonVPN dataset was utilized in the experiment. As previously stated, we have a total of 46,986 sample data. We randomly shuffle the data set and then divide it into training, validation, and test sets in an 8:1:1 ratio. The training and validation sets are primarily utilized for model training, whereas the test set is used to evaluate the model once model training is done.

The model's parameters are as previously stated. The parameters that must be learned are primarily the CNN and Bi-GRU parameters, as well as the weight parameters of the fully connected layer and the output layer. The learning rate, the number of data packets contained in each sample N , the number of bytes contained in each data packet L , and the batch size of the model are all hyperparameters that must be specified. In the experiment, we set the starting learning rate to 0.001 and utilize the Adam optimization technique as the backpropagation optimization algorithm. According to [16], the first few data packets of each flow may be used to complete the classification, hence the experiment took N to be 5, 10, 20, and L to be 64, 128, 256, 512, or 1024. The experimental results reveal that the test set has the greatest influence when N is 5 and L is 1024.

Therefore, the number of data packets N is 5 and the number of bytes is 512. The batch size of the model is set to 64. Table II depicts the model's detailed structure.

TABLE II. DETAILS OF DEEP LEARNING NETWORK MODELS

Model	Architecture Detail
CNN + LSTM	Conv1d(5, 32, 3, 1, 1)-ReLU()-MaxPool(2, 2)-Conv1d(32, 64, 3, 2, 1)-ReLU()-MaxPool(2, 2)-Dropout(0.2)-LSTM(256)-LSTM(128)-Dropout(0.2)-FC(64)-FC(12)
CNN + Bi-LSTM	Conv1d(5, 32, 3, 1, 1)-ReLU()-MaxPool(2, 2)-Conv1d(32, 64, 3, 2, 1)-ReLU()-MaxPool(2, 2)-Dropout(0.2)-Bi-LSTM(128)-Dropout(0.2)-FC(64)-FC(12)
CNN + GRU	Conv1d(5, 32, 3, 1, 1)-ReLU()-MaxPool(2, 2)-Conv1d(32, 64, 3, 2, 1)-ReLU()-MaxPool(2, 2)-Dropout(0.2)-GRU(256)-GRU(128)-Dropout(0.2)-FC(64)-FC(12)
CNN + BiGRU	Conv1d(5, 32, 3, 1, 1)-ReLU()-MaxPool(2, 2)-Conv1d(32, 64, 3, 2, 1)-ReLU()-MaxPool(2, 2)-Dropout(0.2)-Bi-GRU(128)-Dropout(0.2)-FC(64)-FC(12)

2) *Evaluation metrics*: Network traffic classification is fundamentally a multi-classification task, and there are an increasing number of evaluation criteria of classification result, the most frequent of which is accuracy. In addition to the accuracy rate, this paper utilizes the precision rate, recall rate, and F1 score as model evaluation indicators. Assuming N is the total number of samples, TP_c is the number of original category c and predicted as c , FP_c is the number of actual categories that are not category c but the predicted category is category c , TN_c means the actual category is not c and the predicted category is not c , and FN_c means the actual category is c but the expected category is the number of other categories. As a result, the four evaluation indications listed above can be written as follows:

$$Accuracy = \frac{\sum_c TP_c}{N}, P_c = \frac{TP_c}{TP_c + FP_c} \quad (10)$$

$$R_c = \frac{TP_c}{TP_c + FN_c}, F1_c = \frac{2 * P_c * R_c}{P_c + R_c} \quad (11)$$

C. Experimental Result Analysis

It is difficult to compare the algorithms in other articles due to changes in the data collection and experimental setting applied in the study. As a consequence, we chose [8, 11, 12] for comparison, which employed the same dataset as this paper, and the results of the experiment are displayed in Table III.

TABLE III. THE ACCURACY OF DIFFERENT MODELS

	Accuracy
C4.5	0.800
One-dim CNN	0.866
Deep Packet	0.898
Attention Based LSTM	0.912

CNN + LSTM	0.934
CNN + Bi-LSTM	0.932
CNN + GRU	0.928
CNN + Bi-GRU	0.931

The accuracy of the model integrated with CNN and RNN may be increased by roughly 2%, as shown in Table III. The result of combining CNN and LSTM is the best, whereas CNN and Bi-GRU is just 0.3% less effective. According to the findings of the experiments, this combination approach can learn more information in encrypted traffic categorization, making it a useful technique.

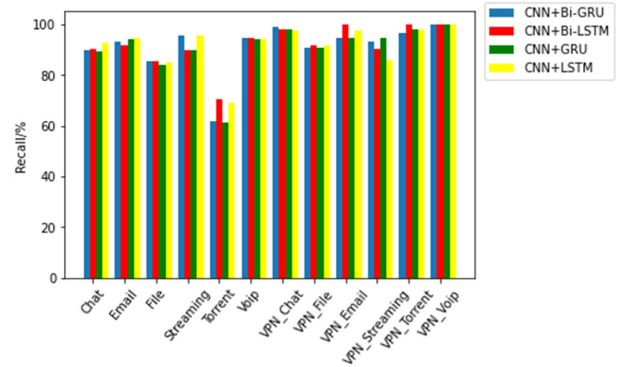


Figure 3. The recall comparison of 4 models

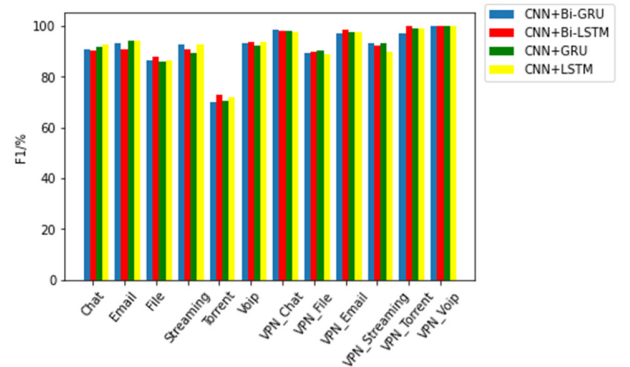


Figure 4. The F1 score comparison of 4 models

TABLE IV. THE COMPARISON OF EFFICIENCY

	weights	Training time(ms)	Test time(ms)
CNN + LSTM	4672k	212	40.6
CNN + Bi-LSTM	4468k	179	36.0
CNN + GRU	3508k	185	36.3
CNN + BiGRU	3387k	166	35.0

The comparisons in Figures 3 and 4 are mostly between different CNN and RNN combinations. GRU's recall and F1 score are tiny in comparison to LSTM, with the exception of the Torrent category, and the other categories are similar or even

greater. Table IV shows that using a bidirectional RNN is superior than using a unidirectional RNN in terms of model parameters and operating efficiency, with Bi-GRU having the shortest parameter scale and the best operational efficiency. The accuracy of the Bi-GRU model is somewhat lower than that of the LSTM model, but the network parameters are 27% lower and the time efficiency is 13.8% quicker. Due to the vast volume of network traffic data, this time benefit will be more apparent in the real-world network environment. As a result, the high criteria for precision in industrial production may be relaxed, and the adoption of Bi-GRU instead of LSTM can drastically cut manufacturing costs.

V. CONCLUSION

In this research, we focus on the categorization of encrypted network traffic, and we suggest a network structure based on CNN and RNN to accomplish the task. Taking into account the temporal and spatial characteristics of network traffic data, we extract the data of the first few packets in the data stream, and then perform data preprocessing through operations such as padding, truncation, and normalization. Convolution is used to learn the spatial characteristics of the data, and Bi-GRU is used to learn the temporal characteristics of the data, reducing the parameter scale of the model while ensuring classification accuracy and increasing classification efficiency. The results of the experiment reveal that CNN and Bi-GRU considerably increased classification and classification efficiency, which is a good notion for dealing with encrypted traffic classification.

In the future, as RNN has some weaknesses in processing network traffic, it will be restricted mostly by sequence length and time efficiency. Our main focus will be on developing a system that can take the place of RNN. Because the attention mechanism and transformer are becoming increasingly common in sequence data, including these structures into the network may have a positive classification impact.

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